

# Fernando Franco Jr.

U.S. Citizen | Active Interim Secret Clearance | Eligible for Top Secret Clearance | LinkedIn: [Fernando Franco Jr](#) | Github: [fernandofrancojr811](#)

## PROFESSIONAL SUMMARY

Software Engineer with experience developing mission-critical aerospace and distributed software systems across **embedded, real-time, cloud, and full-stack environments**. Skilled in **C/C++, Java, Python, Linux** development, software integration, testing, and **Agile SDLC workflows**. Experienced building **telemetry pipelines, REST APIs**, GUI/dashboard systems, and hardware-integrated applications for defense and **AI-focused platforms**. Passionate about scalable aerospace, autonomous, and secure distributed systems.

## EXPERIENCE

**Boeing** | *Software Engineer* | *B-52H Software Blocks Program*

**Dec 2025 - Present**

- Develop and maintain **real-time embedded flight and simulation software** using **C/C++ on Linux-based systems**
- Design and validate hardware-software interfaces supporting mission-critical avionics and simulation environments
- Perform **unit testing, integration testing, and system verification (V&V)** to ensure reliability in mission-critical systems
- Support **Hardware-in-the-Loop (HIL)** and simulation-based testing, debugging and resolving issues to improve system performance and stability

**xVector.us** | *Secure AI Software Engineer*

**Jul 2025 - Sept 2025**

- Developed optimized **C/C++ software for resource-constrained embedded systems**, reducing latency by **50%** and memory usage by **41%**
- Implemented **performance-critical real-time processing components** for AI/ML systems on constrained hardware
- Built **real-time telemetry pipelines** for system observability and performance monitoring
- Contributed to secure deployment workflows emphasizing **controlled execution and system reliability**

**Candidate Tools** | *Software Engineer*

**April 2025 - Jul 2025**

- Prototyped **MCU-based device control software** supporting real-time decision-making systems
- Designed and developed **React and Node.js telemetry dashboards**, reducing debugging cycles by **18%**
- Built **REST APIs and logging systems** in **Python** to improve system observability and diagnostics
- Supported development of **hardware-integrated software systems** with real-time data processing

## TECHNICAL SKILLS

**Languages:** C, C++, Python, Java, TypeScript, SQL, Bash

**Embedded/Systems:** Real-time systems, embedded software, hardware-software integration, Linux/Unix development, simulation environments, debugging, performance optimization, fault-tolerant systems

**Cloud & DevOps:** AWS (S3, Lambda, ECS, CloudWatch, Cognito, KMS), Docker, GitHub Actions, GitLab CI/CD

**Networking:** TCP/IP, System monitoring, Unit Testing, Peer Code Reviews, CMake / Make

**Full-Stack:** React, Node.js, REST APIs, data visualization dashboards

**Tools/Collaboration:** Git, Jira, Agile workflows, IBM DOORS, Jenkins, Confluence, Bitbucket

## EDUCATION

**University of Oklahoma**

*Bachelor of Science in Computer Science*

**Norman, Oklahoma**

**GPA 3.41/4.0**